Project: Worked examples
Title: EN 1993-1-1 member checks (basic)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Resistance checks of a steel member of class 1, 2 or 3 cross-section: tension (6.2.3), compression (6.2.4) and flexural buckling about y and z (6.3.1), bending about y and z (6.2.5), shear with the high-shear reduction (6.2.6, 6.2.8), the linear cross-section interaction (6.2.1(7)) and a simplified member interaction (6.3.3 with user-entered k factors). Lateral-torsional buckling and class 4 sections are outside this sheet.

WHAT IT CALCULATES	RELATION
Axial plastic resistance	$N_{pl,Rd} = A\, f_y/\gamma_{M0}$
Bending resistance, plastic modulus for class 1–2 and elastic for class 3	$M_{c,Rd} = W\, f_y/\gamma_{M0}$
Shear resistance	$V_{pl,Rd} = A_v\, f_y/(\sqrt{3}\,\gamma_{M0})$
Bending resistance reduced by high shear, when $V_{Ed} > 0.5\, V_{pl,Rd}$	$\rho = (2V_{Ed}/V_{pl,Rd} - 1)^2$, resistance $\times (1 - \rho)$
Elastic critical load and non-dimensional slenderness	$N_{cr} = \pi^2 EI/L_{cr}^2$; $\bar{\lambda} = \sqrt{(A\, f_y/N_{cr})}$
Buckling reduction factor and buckling resistance	$\Phi = 0.5[1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$; $\chi = 1/(\Phi + \sqrt{(\Phi^2 - \bar{\lambda}^2)}) \le 1$; $N_{b,Rd} = \chi\, A\, f_y/\gamma_{M1}$
Cross-section check, linear interaction	$N_{Ed}/N_{Rd} + M_{y,Ed}/M_{y,Rd} + M_{z,Ed}/M_{z,Rd} \le 1$
Member check, axial force and bending with buckling	$N_{Ed}/N_{b,Rd,min} + k_{yy}\, M_{y,Ed}/M_{y,Rd} + k_{zz}\, M_{z,Ed}/M_{z,Rd} \le 1$

Basis: Tension[1](6.2.3); Compression[1](6.2.4); Bending moment[1](6.2.5); Shear[1](6.2.6); Bending and shear[1](6.2.8); Linear summation of utilisations[1](6.2.1(7)); Uniform members in compression: buckling resistance, buckling curves[1](6.3.1.1, 6.3.1.2); Uniform members in bending and axial compression[1](6.3.3); Imperfection factors for the buckling curves[2](§6.3.1).

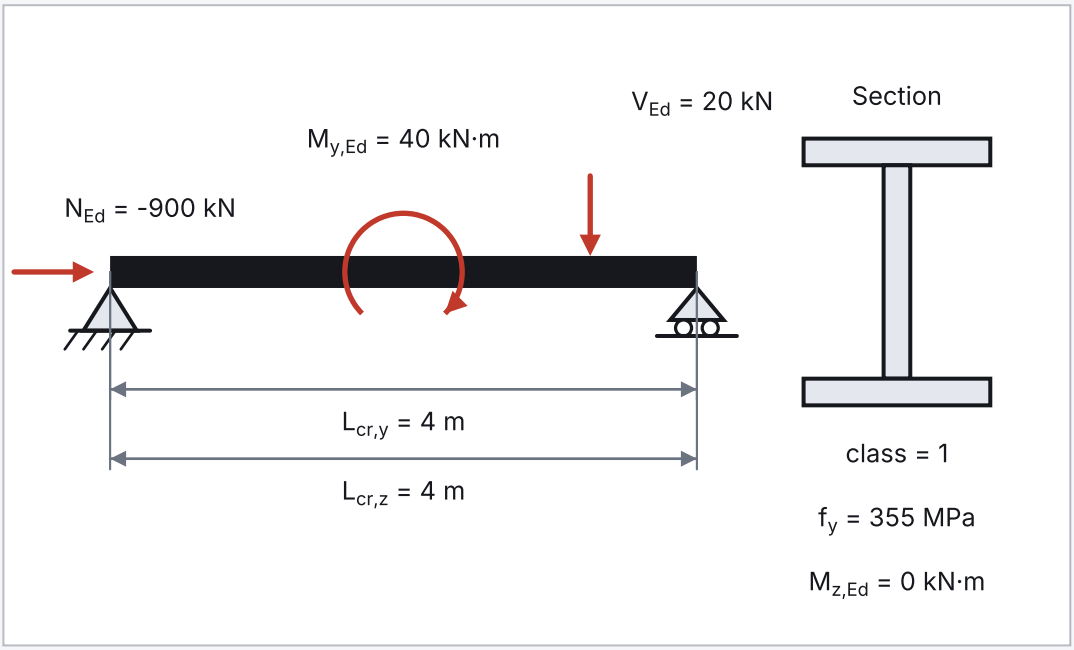


Figure 1 — A steel member under axial force, bending and shear, its buckling lengths about both axes, and its section. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
N_{Ed}	Design axial force, TENSION positive and compression negative (the opposite sign to S6's N_{Ed}, which is compression positive; S2's N_{e1} is tension positive)	-900	kN	reference: S2 frame sheet, column 1, ULS	-100000 … 100000 kN
$M_{y,Ed}$	Design moment about y-y (major)	40	kN·m	reference: S2 frame sheet, column 1, ULS	0 … 100000 kN·m
$M_{z,Ed}$	Design moment about z-z (minor)	0	kN·m	reference: S2 frame sheet	0 … 100000 kN·m
V_{Ed}	Design shear force	20	kN	reference: S2 frame sheet	0 … 100000 kN
f_y	Yield strength (for the thickness)	355	MPa	spec: BS EN 10025-2 S355, t ≤ 16 mm	200 … 700 MPa
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	190 … 220 GPa
γ_{M0}	Partial factor γ_{M0} (cross-section resistance)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
γ_{M1}	Partial factor γ_{M1} (member buckling)	1		spec: UK NA to BS EN 1993-1-1 NA.2.15	1 … 1.25
$section_{class}$	Cross-section class	1		assumption: UC 203×203×46 S355 is class 1 in compression (Blue Book).	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
A	Area A	58.7	cm^2	reference: S1 section sheet / Blue Book UC 203×203×46	1 … 100000 cm^2
$W_{pl,y}$	$W_{pl,y}$	497	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,y}$	$W_{el,y}$	450	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{pl,z}$	$W_{pl,z}$	231	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
$W_{el,z}$	$W_{el,z}$	152	cm^3	reference: S1 section sheet / Blue Book	0.1 … 10000000 cm^3
I_y	I_y	4568	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
I_z	I_z	1548	cm^4	reference: S1 section sheet / Blue Book	0.1 … 1000000000 cm^4
h	Depth h (shear-area helper; leave blank when A_v is given)	203.2	mm	spec: BS EN 10365 UC 203×203×46	1 … 5000 mm
b	Width b (helper)	203.6	mm	spec: BS EN 10365 UC 203×203×46	1 … 5000 mm
t_w	Web thickness (helper)	7.2	mm	spec: BS EN 10365 UC 203×203×46	0.1 … 500 mm
t_f	Flange thickness (helper)	11	mm	spec: BS EN 10365 UC 203×203×46	0.1 … 500 mm
r	Root radius (helper)	10.2	mm	spec: BS EN 10365 UC 203×203×46	0 … 500 mm
η	Shear-area factor η	1		spec: UK NA to BS EN 1993-1-5 NA.2.4	1 … 1.2
$L_{cr,y}$	Buckling length about y-y	4	m	assumption: Pinned both ends: L_cr = L = 4 m.	0.1 … 100 m
$L_{cr,z}$	Buckling length about z-z	4	m	assumption: Pinned both ends: L_cr = L = 4 m.	0.1 … 100 m
$curve_y$	Buckling curve about y-y	b		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, y-y	any

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
$curve_z$	Buckling curve about z-z	c		spec: BS EN 1993-1-1 Table 6.2: rolled, h/b ≤ 1.2, t_f ≤ 100, S355, z-z	any
k_{yy}	Interaction factor k_{yy} (member check)	1		assumption: Simplified interaction: k_yy = 1.0.	0.5 … 2
k_{zz}	Interaction factor k_{zz} (member check)	1		assumption: Simplified interaction: k_zz = 1.0.	0.5 … 2

3. Assumptions

A1. The cross-section class is as stated by the user (1, 2 or 3); class 4 effective properties are not derived here. (6.2.2.5 effective widths are outside this sheet.)
A2. Lateral-torsional buckling is not checked: the compression flange is assumed restrained, or LTB is checked elsewhere. (6.3.2 not implemented.)
A3. The member interaction uses the user's k_{yy} and k_{zz} (default 1.0) with the smaller flexural-buckling reduction on the axial term; the Annex A/B interaction factors are not derived. (6.3.3 simplified.)
A4. Imperfection factors per curve are the textbook values and may be overridden; partial factors default to 1.0 (UK National Annex) and may be overridden. (User-editable with provenance.)
A5. $section_{class}$ (Cross-section class) is taken as 1. (UC 203×203×46 S355 is class 1 in compression (Blue Book).)
A6. $L_{cr,y}$ (Buckling length about y-y) is taken as 4 m. (Pinned both ends: L_{cr} = L = 4 m.)
A7. $L_{cr,z}$ (Buckling length about z-z) is taken as 4 m. (Pinned both ends: L_{cr} = L = 4 m.)
A8. k_{yy} (Interaction factor k_{yy} (member check)) is taken as 1. (Simplified interaction: k_{yy} = 1.0.)
A9. k_{zz} (Interaction factor k_{zz} (member check)) is taken as 1. (Simplified interaction: k_{zz} = 1.0.)

4. Working

[27] $r_h = r$
$= 10.2\ mm$
$= 10.2\ mm$
Root radius used by the helper

[28] $A_{v,rolled} = A - 2 * b * t_f + (t_w + 2 * r_h) * t_f$ [1](6.2.6(3))
$= 58.7\ cm^2 - 2 \times 203.6\ mm \times 11\ mm + (7.2\ mm + 2 \times 10.2\ mm) \times 11\ mm$
$= 16.944\ cm^2$
Rolled I shear area, load parallel to the web

[29] $A_{v,min} = \eta * (h - 2 * t_f) * t_w$ [1](6.2.6(3))
$= 1 \times (203.2\ mm - 2 \times 11\ mm) \times 7.2\ mm$
$= 13.046\ cm^2$
Lower bound $\eta\ h_w\ t_w$

[30] $A_v = max(A_{v,rolled}, A_{v,min})$
$= max(16.944\ cm^2, 13.046\ cm^2)$
$= 16.944\ cm^2$
Shear area

[31] $N_{pl,Rd} = A * f_y / \gamma_{M0}$ [1](6.2.3(2), 6.2.4(2))
$= 58.7\ cm^2 \times 355\ MPa / 1$
$= 2083.8\ kN$
Plastic axial resistance

[32] $W_y = W_{pl,y}$ [1](6.2.5(2))
$= 497\ cm^3$
$= 497\ cm^3$

Section modulus about y (plastic, class 1–2)

[33] $W_z = W_{pl,z}$ [1](6.2.5(2))

$= 231\ cm^3$

$= 231\ cm^3$

Section modulus about z (plastic)

[34] $V_{pl,Rd} = A_v * f_y / (\sqrt{3} * \gamma_{M0})$ [1](6.2.6(2))

$= 16.944\ cm^2 \times 355\ MPa / (\sqrt{3} \times 1)$

$= 347.28\ kN$

Plastic shear resistance

[35] $u_V = V_{Ed} / V_{pl,Rd}$

$= 20\ kN / 347.28\ kN$

$= 0.05759$

Shear utilisation

[36] $\rho = 0$ [1](6.2.8(2))

$= 0$

Shear reduction factor ($V_{Ed} \le 0.5\ V_{pl,Rd}$)

[37] $M_{y,c,Rd} = (1 - \rho) * W_y * f_y / \gamma_{M0}$ [1](6.2.5(2), 6.2.8)

$= (1 - 0) \times 497\ cm^3 \times 355\ MPa / 1$

$= 176.44\ kN \cdot m$

Bending resistance about y

[38] $M_{z,c,Rd} = (1 - \rho) * W_z * f_y / \gamma_{M0}$ [1](6.2.5(2))

$= (1 - 0) \times 231\ cm^3 \times 355\ MPa / 1$

$= 82.005\ kN \cdot m$

Bending resistance about z

[39] $u_{M,y} = M_{y,Ed} / M_{y,c,Rd}$

$= 40\ kN \cdot m / 176.44\ kN \cdot m$

$= 0.22671$

Bending utilisation, y

[40] $u_{M,z} = M_{z,Ed} / M_{z,c,Rd}$

$= 0\ kN \cdot m / 82.005\ kN \cdot m$

$= 0$

Bending utilisation, z

[41] $u_N = \mathrm{abs}(N_{Ed}) / N_{pl,Rd}$

$= \mathrm{abs}(-900\ kN) / 2083.8\ kN$

$= 0.43189$

Compression (cross-section) utilisation

[42] $u_{section} = u_N + u_{M,y} + u_{M,z}$ [1](6.2.1(7))

$= 0.43189 + 0.22671 + 0$

$= 0.65861$

Linear cross-section interaction

[45] α_y = imperfection factor, curve b [1](6.3.1.2)

= Designers' Guide §6.3.1

$= 0.34$

Imperfection factor about y

[46] α_z = imperfection factor, curve c [1](6.3.1.2)

= Designers' Guide §6.3.1

$= 0.49$

Imperfection factor about z

[47] $N_{cr,y} = \mathrm{pi}^2 * E * I_y / L_{cr,y}^2$ [1](6.3.1.2(1))

$= \pi^2 \times 210\ GPa \times 4568\ cm^4 / (4\ m)^2$

$= 5917.3\ kN$

Elastic critical force about y

[48] $\lambda_y = \sqrt{A * f_y / N_{cr,y}}$ [1](6.3.1.2(1))

$= \sqrt{58.7\ cm^2 \times 355\ MPa / 5917.3\ kN}$

$= 0.59343$

Non-dimensional slenderness about y

[49] $\Phi_y = 0.5 * (1 + \alpha_y * (\lambda_y - 0.2) + \lambda_y^2)$ [1](6.3.1.2(1))

$= 0.5 \times (1 + 0.34 \times (0.59343 - 0.2) + 0.59343^2)$

$= 0.74296$

Φ about y

[50] $\chi_y = \min(1, 1 / (\Phi_y + \sqrt{\Phi_y^2 - \lambda_y^2}))$ [1](6.3.1.2(1))

$= \min(1, 1 / (0.74296 + \sqrt{0.74296^2 - 0.59343^2}))$

$= 0.84034$

Reduction factor χ about y

[51] $N_{b,y,Rd} = \chi_y * A * f_y / \gamma_{M1}$ [1](6.3.1.1(3))

$= 0.84034 \times 58.7\ cm^2 \times 355\ MPa / 1$

$= 1751.1\ kN$

Buckling resistance about y

[52] $N_{cr,z} = pi^2 * E * I_z / L_{cr,z}^2$ [1](6.3.1.2(1))

$= \pi^2 \times 210\ GPa \times 1548\ cm^4 / (4\ m)^2$

$= 2005.3\ kN$

Elastic critical force about z

[53] $\lambda_z = \sqrt{A * f_y / N_{cr,z}}$ [1](6.3.1.2(1))

$= \sqrt{58.7\ cm^2 \times 355\ MPa / 2005.3\ kN}$

$= 1.0194$

Non-dimensional slenderness about z

[54] $\Phi_z = 0.5 * (1 + \alpha_z * (\lambda_z - 0.2) + \lambda_z^2)$ [1](6.3.1.2(1))

$= 0.5 \times (1 + 0.49 \times (1.0194 - 0.2) + 1.0194^2)$

$= 1.2204$

Φ about z

[55] $\chi_z = \min(1, 1 / (\Phi_z + \sqrt{\Phi_z^2 - \lambda_z^2}))$ [1](6.3.1.2(1))

$= \min(1, 1 / (1.2204 + \sqrt{1.2204^2 - 1.0194^2}))$

$= 0.52876$

Reduction factor χ about z

[56] $N_{b,z,Rd} = \chi_z * A * f_y / \gamma_{M1}$ [1](6.3.1.1(3))

$= 0.52876 \times 58.7\ cm^2 \times 355\ MPa / 1$

$= 1101.9\ kN$

Buckling resistance about z

[57] $N_{b,Rd} = \min(N_{b,y,Rd}, N_{b,z,Rd})$

$= \min(1751.1\ kN, 1101.9\ kN)$

$= 1101.9\ kN$

Governing buckling resistance

[58] $u_{buckling} = \text{abs}(N_{Ed}) / N_{b,Rd}$ [1](6.3.1.1(1))

$= \text{abs}(-900\ kN) / 1101.9\ kN$

$= 0.8168$

Flexural buckling utilisation

[59] $u_{member} = \text{abs}(N_{Ed}) / N_{b,Rd} + k_{yy} * M_{y,Ed} / M_{y,c,Rd} + k_{zz} * M_{z,Ed} / M_{z,c,Rd}$ [1](6.3.3(4))

$= \text{abs}(-900\ kN) / 1101.9\ kN + 1 \times 40\ kN \cdot m / 176.44\ kN \cdot m + 1 \times 0\ kN \cdot m / 82.005\ kN \cdot m$

$= 1.0435$

Simplified member interaction

5. Results

SYMBOL	VALUE	DESCRIPTION
A_v	16.944 cm²	Shear area
$N_{pl,Rd}$	2083.8 kN	$N_{pl,Rd}$
$V_{pl,Rd}$	347.28 kN	$V_{pl,Rd}$
ρ	0	ρ

SYMBOL	VALUE	DESCRIPTION
$M_{y,c,Rd}$	176.44 kN·m	$M_{y,c,Rd}$
$M_{z,c,Rd}$	82.005 kN·m	$M_{z,c,Rd}$
$N_{cr,y}$	5917.3 kN	$N_{cr,y}$
$N_{cr,z}$	2005.3 kN	$N_{cr,z}$
λ_y	0.59343	$\bar{\lambda}_y$
λ_z	1.0194	$\bar{\lambda}_z$
χ_y	0.84034	χ_y
χ_z	0.52876	χ_z
$N_{b,y,Rd}$	1751.1 kN	$N_{b,y,Rd}$
$N_{b,z,Rd}$	1101.9 kN	$N_{b,z,Rd}$
$N_{b,Rd}$	1101.9 kN	$N_{b,Rd}$
u_N	0.43189	Axial utilisation
$u_{M,y}$	0.22671	Bending y utilisation
$u_{M,z}$	0	Bending z utilisation
u_V	0.05759	Shear utilisation
$u_{section}$	0.65861	Cross-section interaction
$u_{buckling}$	0.8168	Buckling utilisation
u_{member}	1.0435	Member interaction

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Member interaction (simplified)	$u_{member} = 1.0435$	$u_{member} \le 1$	1.04	[1](6.3.3)
PASS	Flexural buckling $N_{Ed} \le N_{b,Rd}$	$u_{buckling} = 0.8168$	$u_{buckling} \le 1$	0.817	[1](6.3.1.1)
PASS	Cross-section interaction	$u_{section} = 0.65861$	$u_{section} \le 1$	0.659	[1](6.2.1(7))
PASS	Compression $N_{Ed} \le N_{c,Rd}$	$u_N = 0.43189$	$u_N \le 1$	0.432	[1](6.2.4)
PASS	Bending y $M_{y,Ed} \le M_{y,c,Rd}$	$u_{M,y} = 0.22671$	$u_{M,y} \le 1$	0.227	[1](6.2.5)
PASS	Shear $V_{Ed} \le V_{pl,Rd}$	$u_V = 0.05759$	$u_V \le 1$	0.0576	[1](6.2.6)
PASS	Bending z $M_{z,Ed} \le M_{z,c,Rd}$	$u_{M,z} = 0$	$u_{M,z} \le 1$	0	[1](6.2.5)

7. Applicability

WARNING — no-ltb: $M_{y,Ed}$: lateral-torsional buckling is not checked by this sheet; restrain the compression flange or check 6.3.2 separately.
WARNING — k-default: k_{yy}, k_{zz}: k_{yy} is left at 1.0 although $N_{Ed}/N_{b,Rd}$ = 0.817 exceeds 0.3. Annex A or B give k above 1 here (at most $C_m(1 + 0.8n)$, which is k_{yy} = 1.411 with C_m = 1); with those values the member interaction would be 1.137 against 1.044 shown. Enter k from Annex A or B.

8. Verification

```
Template struct.en1993-member@1.2.0 Content hash 5c47accf06a9faee Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-1: General Rules and Rules for Buildings. BS EN 1993-1-1. BSI.
2. Gardner L, Nethercot DA. Designers' Guide to EN 1993-1-1: Eurocode 3: Design of Steel Structures, General Rules and Rules for Buildings. Thomas Telford; 2005.